Complete Explicit-Index Derivations for Lecture 7–9: Symmetries and Fields (Poincaré and Lorentz Group Representations)

Global Conventions Established at the Outset

1. **Spacetime Metric**: Minkowski spacetime $\mathbb{R}^{1,3}$ uses the mostly-plus metric signature:

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1), \qquad \eta^{\mu\nu} = \text{diag}(-1, +1, +1, +1), \tag{1}$$

   with Greek indices $\mu, \nu, \rho, \sigma \in \{0, 1, 2, 3\}$ and Latin spatial indices $i, j, k \in \{1, 2, 3\}$. For spatial components, raising or lowering an index does not change the sign ($\eta_{ij} = \delta_{ij}$), whereas raising or lowering a 0-index introduces a minus sign ($\eta_{00} = -1$).

2. **Kronecker Delta**: $\delta^\mu{}_\nu = \eta^{\mu\rho}\eta_{\rho\nu}$ has value 1 if $\mu = \nu$ and 0 otherwise.

3. **3D Levi-Civita Symbol**: $\epsilon_{ijk} = \epsilon^{ijk}$ is completely antisymmetric with $\epsilon_{123} = +1$. It satisfies the contraction identities:

$$\epsilon_{ijk}\epsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}, \qquad \epsilon_{ijk}\epsilon_{ijn} = 2\delta_{kn}. \tag{2}$$

4. **4D Levi-Civita Tensor**: We choose the convention $\epsilon^{0123} = -1$ (so $\epsilon_{0123} = +1$) so that $\epsilon^{0ijk} = -\epsilon_{ijk}$.

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1 Lorentz Group and the $\mathfrak{so}(1,3)$ Lie Algebra (Section 4.1.1)

1.1 Vector Representation and Infinitesimal Generators (Eqs. 4.1 – 4.9)

A Lorentz transformation acts linearly on spacetime coordinates x^μ as:

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu. \tag{3}$$

Preservation of the Minkowski inner product $x'^\mu\eta_{\mu\nu}x'^\nu = x^\rho\eta_{\rho\sigma}x^\sigma$ for arbitrary x gives the defining condition of $O(1,3)$ in explicit index notation (avoiding the matrix shorthand $\Lambda^T\eta\Lambda = \eta$ in Eq. 4.6):

$$\Lambda^\mu{}_\rho\eta_{\mu\nu}\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}. \tag{4}$$

Multiplying both sides by $\eta^{\sigma\tau}(\Lambda^{-1})^\lambda{}_\rho$ yields the inverse transformation rule:

$$(\Lambda^{-1})^\tau{}_\nu = \eta^{\tau\sigma}\eta_{\nu\mu}\Lambda^\mu{}_\sigma = \Lambda_\nu{}^\tau. \tag{5}$$

Convention Choice (Matrix Entries of $l^{\mu\nu}$): For each antisymmetric pair of indices $[\mu\nu]$, the generator $l^{\mu\nu} = -l^{\nu\mu}$ is a linear operator on $\mathbb{R}^{1,3}$ whose components are $(l^{\mu\nu})^{\alpha}{}_{\beta}$, where α is the row (upper) index and β is the column (lower) index. In explicit index notation, the exponential map (Eq. 4.7) is:

$$\Lambda^{\alpha}{}_{\beta} = \left(\exp\left[-\frac{i}{2}\omega_{\mu\nu}l^{\mu\nu}\right]\right)^{\alpha}{}_{\beta} = \delta^{\alpha}{}_{\beta} - \frac{i}{2}\omega_{\mu\nu}(l^{\mu\nu})^{\alpha}{}_{\beta} + \mathcal{O}(\omega^2), \tag{6}$$

where $\omega_{\mu\nu} = -\omega_{\nu\mu}$. Substituting this into the invariance condition $\Lambda^{\gamma}{}_{\alpha}\eta_{\gamma\delta}\Lambda^{\delta}{}_{\beta} = \eta_{\alpha\beta}$ to first order in ω:

$$\begin{aligned}\eta_{\alpha\beta} &= \left(\delta^{\gamma}{}_{\alpha} - \frac{i}{2}\omega_{\mu\nu}(l^{\mu\nu})^{\gamma}{}_{\alpha}\right)\eta_{\gamma\delta}\left(\delta^{\delta}{}_{\beta} - \frac{i}{2}\omega_{\mu\nu}(l^{\mu\nu})^{\delta}{}_{\beta}\right) + \mathcal{O}(\omega^2) \\ &= \eta_{\alpha\beta} - \frac{i}{2}\omega_{\mu\nu}\left[\eta_{\gamma\beta}(l^{\mu\nu})^{\gamma}{}_{\alpha} + \eta_{\alpha\delta}(l^{\mu\nu})^{\delta}{}_{\beta}\right] + \mathcal{O}(\omega^2).\end{aligned} \tag{7}$$

Defining the fully lowered generator matrix $(l^{\mu\nu})_{\alpha\beta} \equiv \eta_{\alpha\gamma}(l^{\mu\nu})^{\gamma}{}_{\beta}$, this requires $(l^{\mu\nu})_{\alpha\beta} = -(l^{\mu\nu})_{\beta\alpha}$. To match the Poincaré algebra (4.4) and the explicit matrices (4.9), the unique choice of basis generators is:

$$(l^{\mu\nu})^{\alpha}{}_{\beta} = i\left(\eta^{\mu\alpha}\delta^{\nu}{}_{\beta} - \eta^{\nu\alpha}\delta^{\mu}{}_{\beta}\right). \tag{8}$$

Verification of Eq. (4.9):

- For $\mu = 0, \nu = 1$:

$$(l^{01})^{\alpha}{}_{\beta} = i\left(\eta^{0\alpha}\delta^{1}{}_{\beta} - \eta^{1\alpha}\delta^{0}{}_{\beta}\right). \tag{9}$$

  Since $\eta^{00} = -1$ and $\eta^{11} = +1$:

$$(l^{01})^{0}{}_{1} = i(-1)(1) = -i, \qquad (l^{01})^{1}{}_{0} = i(0 - (1)(1)) = -i. \tag{10}$$

  Typo Alert in Eq. (4.9): The lecture notes write $l^{01} = +i\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$. However, with $\eta^{00} = -1$, $(l^{01})^{\alpha}{}_{\beta}$ satisfying the Lorentz algebra (4.4) with $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$ must have $-i$ in the $(0,1)$ and $(1,0)$ entries (or equivalently, $+i$ corresponds to $l_{01} = \eta_{00}\eta_{11}l^{01} = -l^{01}$).

- For $\mu = 1, \nu = 2$:

$$(l^{12})^{\alpha}{}_{\beta} = i\left(\eta^{1\alpha}\delta^{2}{}_{\beta} - \eta^{2\alpha}\delta^{1}{}_{\beta}\right) \implies (l^{12})^{1}{}_{2} = +i = -i(-1), \quad (l^{12})^{2}{}_{1} = -i = +i(-1), \tag{11}$$

  which gives:

$$(l^{12})^{\alpha}{}_{\beta} = -i\begin{pmatrix}0&0&0&0\\0&0&-1&0\\0&1&0&0\\0&0&0&0\end{pmatrix}^{\alpha}{}_{\beta}. \tag{12}$$

Explicit Index Derivation of the Lorentz Commutation Relations (Eq. 4.4): Compute the commutator $[l^{\mu\nu}, l^{\rho\sigma}]^{\alpha}{}_{\beta} = (l^{\mu\nu})^{\alpha}{}_{\gamma}(l^{\rho\sigma})^{\gamma}{}_{\beta} - (l^{\rho\sigma})^{\alpha}{}_{\gamma}(l^{\mu\nu})^{\gamma}{}_{\beta}$:

$$\begin{aligned}(l^{\mu\nu})^{\alpha}{}_{\gamma}(l^{\rho\sigma})^{\gamma}{}_{\beta} &= i^2\left(\eta^{\mu\alpha}\delta^{\nu}{}_{\gamma} - \eta^{\nu\alpha}\delta^{\mu}{}_{\gamma}\right)\left(\eta^{\rho\gamma}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\gamma}\delta^{\rho}{}_{\beta}\right) \\ &= -\left(\eta^{\mu\alpha}\eta^{\rho\nu}\delta^{\sigma}{}_{\beta} - \eta^{\mu\alpha}\eta^{\sigma\nu}\delta^{\rho}{}_{\beta} - \eta^{\nu\alpha}\eta^{\rho\mu}\delta^{\sigma}{}_{\beta} + \eta^{\nu\alpha}\eta^{\sigma\mu}\delta^{\rho}{}_{\beta}\right).\end{aligned} \tag{13}$$

Swapping $(\mu,\nu) \leftrightarrow (\rho,\sigma)$ gives the second term:

$$(l^{\rho\sigma})^{\alpha}{}_{\gamma}(l^{\mu\nu})^{\gamma}{}_{\beta} = -\left(\eta^{\rho\alpha}\eta^{\mu\sigma}\delta^{\nu}{}_{\beta} - \eta^{\rho\alpha}\eta^{\nu\sigma}\delta^{\mu}{}_{\beta} - \eta^{\sigma\alpha}\eta^{\mu\rho}\delta^{\nu}{}_{\beta} + \eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta}\right). \tag{14}$$

Subtracting the two and grouping by $\eta^{\nu\rho}, \eta^{\nu\sigma}, \eta^{\mu\sigma}, \eta^{\mu\rho}$:

$$\begin{aligned}[l^{\mu\nu}, l^{\rho\sigma}]^{\alpha}{}_{\beta} &= \eta^{\nu\rho}\left(\eta^{\sigma\alpha}\delta^{\mu}{}_{\beta} - \eta^{\mu\alpha}\delta^{\sigma}{}_{\beta}\right) - \eta^{\nu\sigma}\left(\eta^{\rho\alpha}\delta^{\mu}{}_{\beta} - \eta^{\mu\alpha}\delta^{\rho}{}_{\beta}\right) \\ &\quad + \eta^{\mu\sigma}\left(\eta^{\rho\alpha}\delta^{\nu}{}_{\beta} - \eta^{\nu\alpha}\delta^{\rho}{}_{\beta}\right) - \eta^{\mu\rho}\left(\eta^{\sigma\alpha}\delta^{\nu}{}_{\beta} - \eta^{\nu\alpha}\delta^{\sigma}{}_{\beta}\right) \\ &= i\left[\eta^{\nu\rho}(l^{\mu\sigma})^{\alpha}{}_{\beta} - \eta^{\nu\sigma}(l^{\mu\rho})^{\alpha}{}_{\beta} + \eta^{\mu\sigma}(l^{\nu\rho})^{\alpha}{}_{\beta} - \eta^{\mu\rho}(l^{\nu\sigma})^{\alpha}{}_{\beta}\right]. \end{aligned} \tag{15}$$

This reproduces Eq. (4.4).

Derivation of $[L^{\mu\nu}, P^{\sigma}]$ (Eq. 4.3): In the coordinate representation on scalar functions, $P^{\sigma} = -i\partial^{\sigma}$ and $L^{\mu\nu} = x^{\mu}P^{\nu} - x^{\nu}P^{\mu} = -i(x^{\mu}\partial^{\nu} - x^{\nu}\partial^{\mu})$. Acting on a test function $f(x)$:

$$\begin{aligned}[L^{\mu\nu}, P^{\sigma}]f &= [-i(x^{\mu}\partial^{\nu} - x^{\nu}\partial^{\mu}), -i\partial^{\sigma}]\, f \\ &= -\left(x^{\mu}\partial^{\nu}\partial^{\sigma}f - x^{\nu}\partial^{\mu}\partial^{\sigma}f - \partial^{\sigma}(x^{\mu}\partial^{\nu}f - x^{\nu}\partial^{\mu}f)\right) \\ &= (\partial^{\sigma}x^{\mu})\partial^{\nu}f - (\partial^{\sigma}x^{\nu})\partial^{\mu}f \\ &= \eta^{\sigma\mu}\partial^{\nu}f - \eta^{\sigma\nu}\partial^{\mu}f = i\left(\eta^{\sigma\mu}P^{\nu} - \eta^{\sigma\nu}P^{\mu}\right)f = i\left(P^{\mu}\eta^{\nu\sigma} - P^{\nu}\eta^{\mu\sigma}\right)f, \end{aligned} \tag{16}$$

which matches Eq. (4.3).

1.2 Rotations, Boosts, and the Typo in j_i (Eqs. 4.10 – 4.12)

Typo Resolution in Eq. (4.10): The lecture notes write $j_i = \frac{1}{2}\epsilon_{ijk}l^{jk}$ and $k_i = l^{0i}$. With $\eta_{ij} = +\delta_{ij}$ and the Lorentz algebra (4.4), j_i **must have a minus sign in front**:

$$j_i = -\frac{1}{2}\epsilon_{ijk}l^{jk} \iff l^{mn} = -\epsilon_{mnk}j_k, \qquad k_i = l^{0i}. \tag{17}$$

Let us check the commutator $[j_i, j_j]$ using Eq. (4.4) with $\eta^{mn} = +\delta^{mn}$:

$$\begin{aligned}[j_i, j_j] &= \frac{1}{4}\epsilon_{imn}\epsilon_{jpq}[l^{mn}, l^{pq}] \\ &= \frac{i}{4}\epsilon_{imn}\epsilon_{jpq}\left(\delta^{np}l^{mq} - \delta^{nq}l^{mp} + \delta^{mq}l^{np} - \delta^{mp}l^{nq}\right) \\ &= \frac{i}{4}\epsilon_{imn}\epsilon_{jpq}\left(4\delta^{np}l^{mq}\right) = i\epsilon_{imn}\epsilon_{jnq}l^{mq} \\ &= i\epsilon_{nim}\epsilon_{njq}\left(-\epsilon_{mqk}j_k\right) = -i\left(\delta_{ij}\delta_{mq} - \delta_{iq}\delta_{mj}\right)\epsilon_{mqk}j_k = +i\epsilon_{ijk}j_k. \end{aligned} \tag{18}$$

If $j_i = +\frac{1}{2}\epsilon_{ijk}l^{jk}$ were used without the minus sign, one would obtain $[j_i, j_j] = -i\epsilon_{ijk}j_k$, violating Eq. (4.12).

Hermitian Conjugation Properties (Eq. 4.11): In explicit index notation, the matrix adjoint is $((l^{\mu\nu})^{\dagger})^{\alpha}{}_{\beta} \equiv ((l^{\mu\nu})^{\beta}{}_{\alpha})^{*}$:

- For spatial indices $j, k \in \{1,2,3\}$, $\eta^{j\alpha} = \delta^{j\alpha}$, so:

$$\begin{aligned}\left((j_i)^{\dagger}\right)^{\alpha}{}_{\beta} &= \left((j_i)^{\beta}{}_{\alpha}\right)^{*} = -\frac{1}{2}\epsilon_{ijk}\left[i\left(\delta^{j\beta}\delta^{k}{}_{\alpha} - \delta^{k\beta}\delta^{j}{}_{\alpha}\right)\right]^{*} \\ &= +\frac{i}{2}\epsilon_{ijk}\left(\delta^{j\beta}\delta^{k}{}_{\alpha} - \delta^{k\beta}\delta^{j}{}_{\alpha}\right) = -\frac{i}{2}\epsilon_{ijk}\left(\delta^{j\alpha}\delta^{k}{}_{\beta} - \delta^{k\alpha}\delta^{j}{}_{\beta}\right) = (j_i)^{\alpha}{}_{\beta}. \end{aligned} \tag{19}$$

  Hence $j_i^{\dagger} = j_i$.

- For boost generators $k_i = l^{0i}$, the nonzero entries have $\{\alpha, \beta\} = \{0, i\}$:

$$(k_i)^0{}_i = i(\eta^{00}\delta^i{}_i - \eta^{i0}\delta^0{}_i) = -i, \qquad (k_i)^i{}_0 = i(\eta^{0i}\delta^i{}_0 - \eta^{ii}\delta^0{}_0) = -i. \tag{20}$$

  Taking the conjugate transpose:

$$\left((k_i)^\dagger\right)^0{}_i = \left((k_i)^i{}_0\right)^* = (-i)^* = +i = -(k_i)^0{}_i, \qquad \left((k_i)^\dagger\right)^i{}_0 = \left((k_i)^0{}_i\right)^* = +i = -(k_i)^i{}_0. \tag{21}$$

  Hence $k_i^\dagger = -k_i$.

Derivation of $[j_i, k_j]$ and $[k_i, k_j]$ (Eq. 4.12): Using Eq. (4.4) with $\eta^{00} = -1$, $\eta^{0m} = 0$, and $\eta^{mn} = \delta_{mn}$:

1. **Commutator $[j_i, k_j]$:**

$$\begin{aligned}
[j_i, k_j] &= -\frac{1}{2}\epsilon_{imn}[l^{mn}, l^{0j}] \\
&= -\frac{i}{2}\epsilon_{imn}\left(\eta^{n0}l^{mj} - \eta^{nj}l^{m0} + \eta^{mj}l^{n0} - \eta^{m0}l^{nj}\right) \\
&= -\frac{i}{2}\epsilon_{imn}\left(-\delta_{nj}(-l^{0m}) + \delta_{mj}(-l^{0n})\right) \\
&= -\frac{i}{2}\left(\epsilon_{imj}k_m - \epsilon_{ijn}k_n\right) = i\epsilon_{ijk}k_k.
\end{aligned} \tag{22}$$

2. **Commutator $[k_i, k_j]$:**

$$\begin{aligned}
[k_i, k_j] = [l^{0i}, l^{0j}] &= i\left(\eta^{i0}l^{0j} - \eta^{ij}l^{00} + \eta^{0j}l^{i0} - \eta^{00}l^{ij}\right) \\
&= i\left(-(-1)l^{ij}\right) = il^{ij} = -i\epsilon_{ijk}j_k.
\end{aligned} \tag{23}$$

All three relations in Eq. (4.12) are thus established:

$$[j_i, j_j] = i\epsilon_{ijk}j_k, \qquad [j_i, k_j] = i\epsilon_{ijk}k_k, \qquad [k_i, k_j] = -i\epsilon_{ijk}j_k. \tag{24}$$

1.3 Chiral Decomposition $\mathfrak{so}(1,3)_\mathbb{C} \simeq \mathfrak{sl}(2,\mathbb{C})_L \oplus \mathfrak{sl}(2,\mathbb{C})_R$ (Eqs. 4.13 – 4.24)

Define the complexified combinations (Eq. 4.14):

$$A_i = \frac{1}{2}(j_i + ik_i), \qquad B_i = \frac{1}{2}(j_i - ik_i). \tag{25}$$

Their commutators (Eq. 4.15) follow directly from Eq. (4.12):

$$\begin{aligned}
[A_i, A_j] = \frac{1}{4}[j_i + ik_i, j_j + ik_j] &= \frac{1}{4}\left([j_i, j_j] + i[j_i, k_j] + i[k_i, j_j] - [k_i, k_j]\right) \\
&= \frac{1}{4}\left(i\epsilon_{ijk}j_k + i(i\epsilon_{ijk}k_k) + i(i\epsilon_{ijk}k_k) - (-i\epsilon_{ijk}j_k)\right) \\
&= \frac{1}{2}i\epsilon_{ijk}(j_k + ik_k) = i\epsilon_{ijk}A_k,
\end{aligned} \tag{26}$$

$$\begin{aligned}
[B_i, B_j] = \frac{1}{4}[j_i - ik_i, j_j - ik_j] &= \frac{1}{4}\left([j_i, j_j] - i[j_i, k_j] - i[k_i, j_j] - [k_i, k_j]\right) \\
&= \frac{1}{4}\left(i\epsilon_{ijk}j_k - i(i\epsilon_{ijk}k_k) - i(i\epsilon_{ijk}k_k) - (-i\epsilon_{ijk}j_k)\right) \\
&= \frac{1}{2}i\epsilon_{ijk}(j_k - ik_k) = i\epsilon_{ijk}B_k,
\end{aligned} \tag{27}$$

$$\begin{aligned}
[A_i, B_j] = \frac{1}{4}[j_i + ik_i, j_j - ik_j] &= \frac{1}{4}\left([j_i, j_j] - i[j_i, k_j] + i[k_i, j_j] + [k_i, k_j]\right) \\
&= \frac{1}{4}\left(i\epsilon_{ijk}j_k - i(i\epsilon_{ijk}k_k) + i(i\epsilon_{ijk}k_k) + (-i\epsilon_{ijk}j_k)\right) = 0.
\end{aligned} \tag{28}$$

Reality Condition (Eq. 4.21): Following the lecture notes' convention where complex conjugation $*$ on the complexified algebra reverses the i in front of the real Lie algebra generators $(-ij_i)^* = -ij_i \implies j_i^* = -j_i$ and $(-ik_i)^* = -ik_i \implies k_i^* = -k_i$:

$$A_i^* = \left(\frac{1}{2}(j_i + ik_i)\right)^* = \frac{1}{2}\left(j_i^* - ik_i^*\right) = \frac{1}{2}\left(-j_i + ik_i\right) = -B_i, \tag{29}$$

which gives Eq. (4.21), and implies that complex conjugation of a representation swaps the (j_L, j_R) labels: $(j_L, j_R)^* = (j_R, j_L)$ (Eq. 4.23).

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2 Spinors, $SL(2,\mathbb{C})$, and Index Conventions (Section 4.1.2)

2.1 Index Structure of $\epsilon_{\alpha\beta}$, $\epsilon^{\alpha\beta}$, and Spinor Raising/Lowering (Eqs. 4.47 – 4.71)

Convention Choices (Professor's Convention $\epsilon_{\dot\alpha\dot\beta} = -\epsilon_{\alpha\beta}$):

- Undotted indices $\alpha, \beta, \gamma, \delta \in \{1, 2\}$ label the fundamental $(\frac{1}{2}, 0)$ representation of $SL(2,\mathbb{C})$.
- Dotted indices $\dot\alpha, \dot\beta, \dot\gamma, \dot\delta \in \{1, 2\}$ label the complex-conjugate $(0, \frac{1}{2})$ representation of $SL(2,\mathbb{C})$.
- **Antisymmetric Spinor Metric (Eq. 4.55 & Professor's Dotted Convention):**

$$\epsilon_{12} = +1, \quad \epsilon_{21} = -1, \quad \epsilon_{11} = \epsilon_{22} = 0; \qquad \epsilon^{12} = -1, \quad \epsilon^{21} = +1, \quad \epsilon^{11} = \epsilon^{22} = 0, \tag{30}$$

$$\epsilon_{\dot1\dot2} = -1, \quad \epsilon_{\dot2\dot1} = +1, \quad \epsilon_{\dot1\dot1} = \epsilon_{\dot2\dot2} = 0; \qquad \epsilon^{\dot1\dot2} = +1, \quad \epsilon^{\dot2\dot1} = -1, \quad \epsilon^{\dot1\dot1} = \epsilon^{\dot2\dot2} = 0. \tag{31}$$

  Note that because $\epsilon_{\dot\alpha\dot\beta} = -\epsilon_{\alpha\beta}$ and $\epsilon^{\dot\alpha\dot\beta} = -\epsilon^{\alpha\beta}$, the diagram of index-raising and Hermitian conjugation (†) anticommutes: if $(\psi_\alpha)^\dagger = \bar\psi_{\dot\alpha}$, then:

$$(\psi^\alpha)^\dagger = (\epsilon^{\alpha\beta}\psi_\beta)^\dagger = \epsilon^{\alpha\beta}\bar\psi_{\dot\beta} = -\epsilon^{\dot\alpha\dot\beta}\bar\psi_{\dot\beta} = -\bar\psi^{\dot\alpha}. \tag{32}$$

Verification of Eq. (4.56) and (4.70): Contracting the second index of $\epsilon^{\alpha\beta}$ with the first index of $\epsilon_{\beta\gamma}$ (and identically for dotted indices, since both $\epsilon^{\dot\alpha\dot\beta}$ and $\epsilon_{\dot\beta\dot\gamma}$ flip sign):

$$\epsilon^{\alpha\beta}\epsilon_{\beta\gamma} = \delta^\alpha{}_\gamma, \qquad \epsilon^{\alpha\beta}\epsilon_{\gamma\beta} = -\delta^\alpha{}_\gamma, \qquad \epsilon^{\dot\alpha\dot\beta}\epsilon_{\dot\beta\dot\gamma} = \delta^{\dot\alpha}{}_{\dot\gamma}, \qquad \epsilon^{\dot\alpha\dot\beta}\epsilon_{\dot\gamma\dot\beta} = -\delta^{\dot\alpha}{}_{\dot\gamma}. \tag{33}$$

Consistency of Raising and Lowering (Eq. 4.57): Given $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta$ and $\psi_\alpha = \epsilon_{\alpha\beta}\psi^\beta$ (and $\bar\chi^{\dot\alpha} = \epsilon^{\dot\alpha\dot\beta}\bar\chi_{\dot\beta}$, $\bar\chi_{\dot\alpha} = \epsilon_{\dot\alpha\dot\beta}\bar\chi^{\dot\beta}$), raising and then lowering returns the original spinor:

$$\epsilon_{\alpha\beta}\psi^\beta = \epsilon_{\alpha\beta}\epsilon^{\beta\gamma}\psi_\gamma = \delta_\alpha{}^\gamma\psi_\gamma = \psi_\alpha, \qquad \epsilon_{\dot\alpha\dot\beta}\bar\chi^{\dot\beta} = \epsilon_{\dot\alpha\dot\beta}\epsilon^{\dot\beta\dot\gamma}\bar\chi_{\dot\gamma} = \delta_{\dot\alpha}{}^{\dot\gamma}\bar\chi_{\dot\gamma} = \bar\chi_{\dot\alpha}. \tag{34}$$

Derivation of $SL(2,\mathbb{C})$ Invariance of $\epsilon_{\alpha\beta}$ (Eq. 4.59) and Dual Transformation (Eqs. 4.60 – 4.68): Let an $SL(2,\mathbb{C})$ matrix have components $R_\alpha{}^\beta$ (where α is the row index and β is the column index, as in Eqs. 4.48 and 4.60). In explicit index notation, $(R^T\epsilon R)_{\alpha\beta}$ is:

$$R_\gamma{}^\alpha\epsilon_{\gamma\delta}R_\delta{}^\beta = R_1{}^\alpha\epsilon_{12}R_2{}^\beta + R_2{}^\alpha\epsilon_{21}R_1{}^\beta = R_1{}^\alpha R_2{}^\beta - R_2{}^\alpha R_1{}^\beta = (\det R)\epsilon_{\alpha\beta} = \epsilon_{\alpha\beta}. \tag{35}$$

Similarly, the cofactor formula for the 2×2 inverse of a unimodular matrix ($\det R = 1$) gives Eq. (4.65):

$$\epsilon^{\alpha\beta}R_\beta{}^\gamma\epsilon_{\gamma\delta} = (R^{-1})_\delta{}^\alpha. \tag{36}$$

Substituting this into Eq. (4.62):

$$\psi'^{\alpha} = \epsilon^{\alpha\beta}\psi'_{\beta} = \epsilon^{\alpha\beta}R_{\beta}{}^{\gamma}\psi_{\gamma} = \epsilon^{\alpha\beta}R_{\beta}{}^{\gamma}\epsilon_{\gamma\delta}\psi^{\delta} = (R^{-1})_{\delta}{}^{\alpha}\psi^{\delta} = (R^{-1})_{\beta}{}^{\alpha}\psi^{\beta}, \tag{37}$$

which proves Eq. (4.66). Checking invariance of the spinor inner product $\psi^{\alpha}\chi_{\alpha}$ (Eqs. 4.67–4.68):

$$\psi'^{\alpha}\chi'_{\alpha} = (R^{-1})_{\beta}{}^{\alpha}\psi^{\beta}R_{\alpha}{}^{\gamma}\chi_{\gamma} = (R^{-1})_{\beta}{}^{\alpha}R_{\alpha}{}^{\gamma}\psi^{\beta}\chi_{\gamma} = \delta_{\beta}{}^{\gamma}\psi^{\beta}\chi_{\gamma} = \psi^{\beta}\chi_{\beta}. \tag{38}$$

For commuting spinors (Eq. 4.69):

$$\psi^{\alpha}\chi_{\alpha} = \epsilon^{\alpha\beta}\psi_{\beta}\chi_{\alpha} = \epsilon^{\beta\alpha}\psi_{\alpha}\chi_{\beta} = -\epsilon^{\alpha\beta}\psi_{\alpha}\chi_{\beta} = -\psi_{\alpha}\chi^{\alpha}. \tag{39}$$

2.2 Four-Vectors as 2×2 Hermitian Matrices and Spinor Bilinears (Eqs. 4.35 – 4.46, 4.72 – 4.95)

2.2.1 Part A: Derivation of Eqs. (4.35) – (4.46)

With $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$, lowering the index on $x^{\mu} = (x^0, x^1, x^2, x^3)$ gives $x_{\mu} = \eta_{\mu\nu}x^{\nu} = (-x^0, x^1, x^2, x^3)$. With $(\sigma^{\mu})_{\alpha\dot{\beta}} = (\mathbf{1}, \sigma^1, \sigma^2, \sigma^3)_{\alpha\dot{\beta}}$:

$$X_{\alpha\dot{\beta}} = x_{\mu}(\sigma^{\mu})_{\alpha\dot{\beta}} = -x^0\delta_{\alpha\dot{\beta}} + x^i(\sigma_i)_{\alpha\dot{\beta}} = \begin{pmatrix} -x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & -x^0 - x^3 \end{pmatrix}_{\alpha\dot{\beta}}. \tag{40}$$

Computing the determinant (Eqs. 4.36 and 4.78):

$$\det X = (-x^0+x^3)(-x^0-x^3)-(x^1-ix^2)(x^1+ix^2) = (x^0)^2-(x^1)^2-(x^2)^2-(x^3)^2 = -x^{\mu}x_{\mu}. \tag{41}$$

Under $X'_{\alpha\dot{\beta}} = R_{\alpha}{}^{\gamma}X_{\gamma\dot{\delta}}(R^{\dagger})^{\dot{\delta}}{}_{\dot{\beta}}$ where $\det R = 1$, we have $\det X' = (\det R)(\det X)(\det R^{\dagger}) = \det X$, so $-(x')^{\mu}x'_{\mu} = -x^{\mu}x_{\mu}$. For a pure spatial rotation leaving x^0 invariant ($x'^0 = x^0$ for $x^i = 0$), setting $X_{\gamma\dot{\delta}} = -x^0\delta_{\gamma\dot{\delta}}$ requires:

$$R_{\alpha}{}^{\gamma}\delta_{\gamma\dot{\delta}}(R^{\dagger})^{\dot{\delta}}{}_{\dot{\beta}} = \delta_{\alpha\dot{\beta}} \implies RR^{\dagger} = \mathbf{1} \implies R \in SU(2). \tag{42}$$

2.2.2 Part B: Derivation of Eqs. (4.75) – (4.91) and Resolution of Typo in Eq. (4.88)

Following the index convention of pages 10–11:

- ψ_L^{β} carries an upper undotted index and transforms as (Eq. 4.84):

$$\psi_L^{\alpha} \longrightarrow \psi_L'^{\alpha} = R^{\alpha}{}_{\beta}\psi_L^{\beta}. \tag{43}$$

- Its Hermitian conjugate $\psi_L^{\dagger\dot{\alpha}} \equiv (\psi_L^{\alpha})^*$ carries an upper dotted index and transforms as (Eq. 4.85):

$$\psi_L^{\dagger\dot{\alpha}} \longrightarrow \psi_L'^{\dagger\dot{\alpha}} = (R^{\alpha}{}_{\beta})^*(\psi_L^{\beta})^* = \psi_L^{\dagger\dot{\beta}}(R^{\dagger})_{\dot{\beta}}{}^{\dot{\alpha}}, \tag{44}$$

  where $(R^{\dagger})_{\dot{\beta}}{}^{\dot{\alpha}} \equiv (R^{\alpha}{}_{\beta})^*$.

- In Eqs. (4.80)–(4.81), $X_{\dot{\alpha}\beta} = x_{\mu}(\sigma^{\mu})_{\dot{\alpha}\beta}$ transforms as:

$$X'_{\dot{\alpha}\beta} = x'_{\mu}(\sigma^{\mu})_{\dot{\alpha}\beta} = (R^{\dagger})_{\dot{\alpha}}{}^{\dot{\gamma}}X_{\dot{\gamma}\delta}R^{\delta}{}_{\beta} = x_{\nu}(R^{\dagger})_{\dot{\alpha}}{}^{\dot{\gamma}}(\sigma^{\nu})_{\dot{\gamma}\delta}R^{\delta}{}_{\beta}. \tag{45}$$

- Meanwhile, under $x'^{\mu} = \Lambda^{\mu}{}_{\nu}x^{\nu}$, the covector $x_{\mu} = \eta_{\mu\rho}x^{\rho}$ transforms with the inverse Lorentz matrix (Eq. 4.82):

$$x'_{\mu} = x_{\nu}(\Lambda^{-1})^{\nu}{}_{\mu}. \tag{46}$$

  Equating the two expressions for $X'_{\dot{\alpha}\beta}$ for arbitrary x_{ν} yields Eq. (4.83):

$$(R^{\dagger})_{\dot{\alpha}}{}^{\dot{\gamma}}(\sigma^{\mu})_{\dot{\gamma}\delta}R^{\delta}{}_{\beta} = (\Lambda^{-1})^{\mu}{}_{\nu}(\sigma^{\nu})_{\dot{\alpha}\beta}. \tag{47}$$

- Substituting this into the transformation of $V^\mu = \psi_L^{\dagger\dot{\alpha}}(\sigma^\mu)_{\dot{\alpha}\beta}\psi_L^\beta$ (Eqs. 4.86–4.88):

$$\begin{aligned} V^\mu \longrightarrow V'^\mu &= \psi_L'^{\dagger\dot{\alpha}}(\sigma^\mu)_{\dot{\alpha}\beta}\psi_L'^\beta \\ &= \psi_L^{\dagger\dot{\gamma}}(R^\dagger)_{\dot{\gamma}}{}^{\dot{\alpha}}(\sigma^\mu)_{\dot{\alpha}\beta}R^\beta{}_\delta\psi_L^\delta \\ &= \psi_L^{\dagger\dot{\gamma}}\left[(\Lambda^{-1})^\mu{}_\nu(\sigma^\nu)_{\dot{\gamma}\delta}\right]\psi_L^\delta = (\Lambda^{-1})^\mu{}_\nu V^\nu. \end{aligned} \tag{48}$$

Typo Note in Eq. (4.88): Equation (4.88) in the lecture notes writes $\Lambda^\mu{}_\nu$ in the middle expression and $(\Lambda^{-1})^\mu{}_\nu$ on the right-hand side. If $R = \exp(-\frac{i}{2}\theta^i\sigma_i - \frac{1}{2}\eta^i\sigma_i)$ is the active transformation on ψ_L, then $V'^\mu = \Lambda^\mu{}_\nu V^\nu$ (as proven in Exercise 4.1 below), whereas if Λ is defined via $X' = R^\dagger X R$, Λ is replaced by Λ^{-1}.

2.3 Solution to Exercise 4.1 and Exercise 4.2 (Eqs. 4.92 – 4.95)

2.3.1 Exercise 4.1: Explicit Transformation of $V^\mu = (\psi_L^\dagger\psi_L, \psi_L^\dagger\sigma^i\psi_L)$ Under Boosts

On the left-handed Weyl spinor $(\frac{1}{2}, 0)$, $B_i = \frac{1}{2}(j_i - ik_i) = 0$ and $A_i = \frac{1}{2}(j_i + ik_i) = \frac{1}{2}\sigma_i$:

$$j_i = A_i + B_i = \frac{1}{2}\sigma_i, \qquad k_i = -i(A_i - B_i) = -\frac{i}{2}\sigma_i. \tag{49}$$

Under a Lorentz boost with rapidity $\vec{\eta} = \eta\hat{n}$ ($\hat{n}^i\hat{n}^i = 1$), the $SL(2,\mathbb{C})$ matrix acting on ψ_L^α is:

$$R^\alpha{}_\beta = \left(\exp\left(-i\eta^i k_i\right)\right)^\alpha{}_\beta = \left(\exp\left(-\frac{1}{2}\eta\hat{n}^i\sigma_i\right)\right)^\alpha{}_\beta = \cosh\left(\frac{\eta}{2}\right)\delta^\alpha{}_\beta - \sinh\left(\frac{\eta}{2}\right)\hat{n}^i(\sigma_i)^\alpha{}_\beta. \tag{50}$$

Since the Pauli matrices are Hermitian, $((\sigma_i)^\alpha{}_\beta)^* = (\sigma_i)_{\dot{\beta}}{}^{\dot{\alpha}}$, the Hermitian conjugate matrix is:

$$(R^\dagger)_{\dot{\gamma}}{}^{\dot{\alpha}} = \cosh\left(\frac{\eta}{2}\right)\delta_{\dot{\gamma}}{}^{\dot{\alpha}} - \sinh\left(\frac{\eta}{2}\right)\hat{n}^i(\sigma_i)_{\dot{\gamma}}{}^{\dot{\alpha}}. \tag{51}$$

Using the Pauli matrix product identity in explicit index notation:

$$(\sigma_i)_{\dot{\alpha}\beta}(\sigma_j)^\beta{}_\delta = \delta_{ij}\delta_{\dot{\alpha}\delta} + i\epsilon_{ijk}(\sigma_k)_{\dot{\alpha}\delta}, \tag{52}$$

we compute the transformations of V^0 and V^k:

1. **Transformation of V^0:**

$$\begin{aligned} (R^\dagger)_{\dot{\gamma}}{}^{\dot{\alpha}}\delta_{\dot{\alpha}\beta}R^\beta{}_\delta &= \left[\cosh\left(\frac{\eta}{2}\right)\delta_{\dot{\gamma}\beta} - \sinh\left(\frac{\eta}{2}\right)\hat{n}^i(\sigma_i)_{\dot{\gamma}\beta}\right]\left[\cosh\left(\frac{\eta}{2}\right)\delta^\beta{}_\delta - \sinh\left(\frac{\eta}{2}\right)\hat{n}^j(\sigma_j)^\beta{}_\delta\right] \\ &= \cosh^2\left(\frac{\eta}{2}\right)\delta_{\dot{\gamma}\delta} - 2\sinh\left(\frac{\eta}{2}\right)\cosh\left(\frac{\eta}{2}\right)\hat{n}^i(\sigma_i)_{\dot{\gamma}\delta} \\ &\quad + \sinh^2\left(\frac{\eta}{2}\right)\hat{n}^i\hat{n}^j\left(\delta_{ij}\delta_{\dot{\gamma}\delta} + i\epsilon_{ijk}(\sigma_k)_{\dot{\gamma}\delta}\right) \\ &= \cosh(\eta)\delta_{\dot{\gamma}\delta} - \sinh(\eta)\hat{n}^i(\sigma_i)_{\dot{\gamma}\delta}. \end{aligned} \tag{53}$$

   Contracting with $\psi_L^{\dagger\dot{\gamma}}$ and ψ_L^δ yields:

$$V'^0 = \cosh(\eta)V^0 - \sinh(\eta)\hat{n}^i V^i. \tag{54}$$

2. **Transformation of V^k:** First compute $(\sigma^k)_{\dot{\alpha}\beta}R^\beta{}_\delta$:

$$(\sigma^k)_{\dot{\alpha}\beta}R^\beta{}_\delta = \cosh\left(\frac{\eta}{2}\right)(\sigma^k)_{\dot{\alpha}\delta} - \sinh\left(\frac{\eta}{2}\right)\hat{n}^j\left(\delta_{kj}\delta_{\dot{\alpha}\delta} + i\epsilon_{kjm}(\sigma_m)_{\dot{\alpha}\delta}\right). \tag{55}$$

   Now multiply on the left by $(R^\dagger)_{\dot{\gamma}}{}^{\dot{\alpha}}$:

$$(R^\dagger)_{\dot{\gamma}}{}^{\dot{\alpha}}(\sigma^k)_{\dot{\alpha}\beta}R^\beta{}_\delta = \cosh^2\left(\frac{\eta}{2}\right)(\sigma^k)_{\dot{\gamma}\delta} - \cosh\left(\frac{\eta}{2}\right)\sinh\left(\frac{\eta}{2}\right)\hat{n}^j\left(\delta_{kj}\delta_{\dot{\gamma}\delta} + i\epsilon_{kjm}(\sigma_m)_{\dot{\gamma}\delta}\right)$$

$$
\begin{aligned}
&- \sinh\left(\frac{\eta}{2}\right)\cosh\left(\frac{\eta}{2}\right)\hat{n}^i\left(\delta_{ik}\delta_{\dot{\gamma}\delta} + i\epsilon_{ikm}(\sigma_m)_{\dot{\gamma}\delta}\right) \\
&+ \sinh^2\left(\frac{\eta}{2}\right)\hat{n}^i\hat{n}^j\left[\delta_{kj}(\sigma_i)_{\dot{\gamma}\delta} + i\epsilon_{kjm}\left(\delta_{im}\delta_{\dot{\gamma}\delta} + i\epsilon_{imn}(\sigma_n)_{\dot{\gamma}\delta}\right)\right] \\
&= (\sigma^k)_{\dot{\gamma}\delta} - \sinh(\eta)\hat{n}^k\delta_{\dot{\gamma}\delta} + (\cosh(\eta)-1)\hat{n}^k\hat{n}^n(\sigma_n)_{\dot{\gamma}\delta}. \qquad (56)
\end{aligned}
$$

Contracting with $\psi_L^{\dagger\dot{\gamma}}$ and ψ_L^{δ} yields:

$$
V'^k = V^k - \sinh(\eta)\hat{n}^k V^0 + (\cosh(\eta)-1)\hat{n}^k(\hat{n}^n V^n). \qquad (57)
$$

2.3.2 Exercise 4.2: Explicit Transformation of $\bar{V}^\mu = \psi_R^\dagger\bar{\sigma}^\mu\psi_R$ and Origin of $\bar{\sigma}^0 = -\mathbf{1}$

For a right-handed Weyl spinor ψ_R transforming in $(0, \frac{1}{2})$, $A_i = \frac{1}{2}(j_i + ik_i) = 0$ and $B_i = \frac{1}{2}(j_i - ik_i) = \frac{1}{2}\sigma_i$, so:

$$
j_i = A_i + B_i = \frac{1}{2}\sigma_i, \qquad k_i = -i(A_i - B_i) = +\frac{i}{2}\sigma_i. \qquad (58)
$$

Because k_i has the opposite sign ($+\frac{i}{2}\sigma_i$) compared to ψ_L ($-\frac{i}{2}\sigma_i$) while j_i has the same sign, a boost by rapidity $\vec{\eta}$ acts on ψ_R with $\eta \to -\eta$:

$$
(\psi_R^\dagger\mathbf{1}\psi_R)' = \cosh(\eta)(\psi_R^\dagger\mathbf{1}\psi_R) + \sinh(\eta)\hat{n}^i(\psi_R^\dagger\sigma^i\psi_R), \qquad (59)
$$

$$
(\psi_R^\dagger\sigma^k\psi_R)' = (\psi_R^\dagger\sigma^k\psi_R) + \sinh(\eta)\hat{n}^k(\psi_R^\dagger\mathbf{1}\psi_R) + (\cosh(\eta)-1)\hat{n}^k\hat{n}^n(\psi_R^\dagger\sigma^n\psi_R). \qquad (60)
$$

Thus, for $\bar{V}^\mu = (\bar{V}^0, \bar{V}^k)$ to transform with the **same** boost matrix $\Lambda^\mu{}_\nu$ as V^μ, $\bar{V}^0 = -\psi_R^\dagger\mathbf{1}\psi_R$ must have a relative minus sign compared to $\bar{V}^k = +\psi_R^\dagger\sigma^k\psi_R$, i.e., $\bar{\sigma}^\mu = (-\mathbf{1}, \sigma^i)$.

Index/Metric Origin of the Minus Sign in $\bar{\sigma}^0 = -\mathbf{1}$: In spinor index notation with $\epsilon^{\dot{\alpha}\dot{\gamma}} = -\epsilon^{\alpha\gamma}$, raising both indices of $(\sigma^\mu)_{\dot{\gamma}\delta}$ via:

$$
(\bar{\sigma}^\mu)^{\beta\dot{\alpha}} = -\epsilon^{\beta\delta}\epsilon^{\dot{\alpha}\dot{\gamma}}(\sigma^\mu)_{\dot{\gamma}\delta} \qquad (61)
$$

gives:

- For $\mu = 0$, $(\sigma^0)_{\dot{\gamma}\delta} = \delta_{\dot{\gamma}\delta}$, so $-\epsilon^{\beta\delta}\epsilon^{\dot{\alpha}\dot{\gamma}}\delta_{\dot{\gamma}\delta} = -\epsilon^{\beta\delta}(-\epsilon^{\dot{\alpha}\delta}) = -\delta^{\beta\dot{\alpha}} = -(\mathbf{1})^{\beta\dot{\alpha}}$.
- For $\mu = i$, $-\epsilon^{\beta\delta}\epsilon^{\dot{\alpha}\dot{\gamma}}(\sigma^i)_{\dot{\gamma}\delta} = +(\sigma^{i*})^{\beta\dot{\alpha}}$.

Equivalently, from the Clifford identity $(\sigma^\mu)_{\dot{\alpha}\beta}(\bar{\sigma}^\nu)^{\beta\dot{\gamma}} + (\sigma^\nu)_{\dot{\alpha}\beta}(\bar{\sigma}^\mu)^{\beta\dot{\gamma}} = 2\eta^{\mu\nu}\delta_{\dot{\alpha}}{}^{\dot{\gamma}}$ in $(-+++)$ signature, $\eta^{00} = -1$ forces $\sigma^0\bar{\sigma}^0 = -\mathbf{1} \implies \bar{\sigma}^0 = -\mathbf{1}$, while $\eta^{ii} = +1$ forces $\sigma^i\bar{\sigma}^i = +\mathbf{1} \implies \bar{\sigma}^i = +\sigma^i$.

---

3 One-Particle States, Little Group, and Casimir Operators (Section 4.1.3)

3.1 First Casimir Operator $C_1 = -P_\mu P^\mu$ (Eqs. 4.96 – 4.121)

Using $[P^\mu, P^\nu] = 0$ and $[L^{\mu\nu}, P^\rho] = i(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu)$:

$$
\begin{aligned}
[L^{\mu\nu}, -P_\rho P^\rho] &= -\eta_{\rho\sigma}\left([L^{\mu\nu}, P^\rho]P^\sigma + P^\rho[L^{\mu\nu}, P^\sigma]\right) \\
&= -i\eta_{\rho\sigma}\left((\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu)P^\sigma + P^\rho(\eta^{\nu\sigma}P^\mu - \eta^{\mu\sigma}P^\nu)\right) \\
&= -i\left(P^\mu P^\nu - P^\nu P^\mu + P^\nu P^\mu - P^\mu P^\nu\right) = 0. \qquad (62)
\end{aligned}
$$

3.2 Induced Representations and Wigner Rotation (Eqs. 4.122 – 4.128)

For reference momentum k^μ and boost $L(p)^\mu{}_\nu k^\nu = p^\mu$, let $W(\Lambda,p)^\mu{}_\nu \equiv (L^{-1}(\Lambda p))^\mu{}_\rho \Lambda^\rho{}_\sigma L(p)^\sigma{}_\nu$. Acting on k^ν in explicit index notation (Eq. 4.126):

$$\begin{aligned} W(\Lambda,p)^\mu{}_\nu k^\nu &= (L^{-1}(\Lambda p))^\mu{}_\rho \Lambda^\rho{}_\sigma \left(L(p)^\sigma{}_\nu k^\nu\right) \\ &= (L^{-1}(\Lambda p))^\mu{}_\rho \left(\Lambda^\rho{}_\sigma p^\sigma\right) = (L^{-1}(\Lambda p))^\mu{}_\rho (\Lambda p)^\rho = k^\mu, \end{aligned} \tag{63}$$

so $W(\Lambda,p) \in G_k$ (Eq. 4.127), and:

$$U(\Lambda)|p,\lambda\rangle = U(L(\Lambda p))U(W(\Lambda,p))|k,\lambda\rangle = \sum_{\lambda'} |\Lambda p,\lambda'\rangle D_{\lambda'\lambda}(W(\Lambda,p)). \tag{64}$$

3.3 The Pauli-Lubanski Vector and Second Casimir $C_2 = W_\mu W^\mu$ (Eqs. 4.129 – 4.136)

The Pauli-Lubanski vector is defined as (Eq. 4.129):

$$W^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu L_{\rho\sigma}. \tag{65}$$

1. **Proof that $W^\mu P_\mu = 0$ and $[W^\mu, P^\alpha] = 0$ (Eq. 4.130):**

$$\begin{aligned} [W^\mu, P^\alpha] &= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu [L_{\rho\sigma}, P^\alpha] = \frac{i}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu \left(\delta_\sigma{}^\alpha P_\rho - \delta_\rho{}^\alpha P_\sigma\right) \\ &= \frac{i}{2}\left(\epsilon^{\mu\nu\rho\alpha} P_\nu P_\rho - \epsilon^{\mu\nu\alpha\sigma} P_\nu P_\sigma\right) = 0, \end{aligned} \tag{66}$$

$$W^\mu P_\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\mu P_\nu L_{\rho\sigma} = 0. \tag{67}$$

2. **Proof that $[W^\lambda, L^{\mu\nu}] = i(W^\mu \eta^{\lambda\nu} - W^\nu \eta^{\lambda\mu})$ (Eq. 4.131):** Using the 5-index antisymmetrization identity $\epsilon^{[\lambda\tau\beta\gamma}\eta^{\nu]\mu} = 0$ and the Leibniz rule:

$$\begin{aligned} [L^{\mu\nu}, W^\lambda] &= \frac{1}{2}\epsilon^{\lambda\alpha\beta\gamma}\left([L^{\mu\nu}, P_\alpha]L_{\beta\gamma} + P_\alpha [L^{\mu\nu}, L_{\beta\gamma}]\right) \\ &= \frac{i}{2}\left[\epsilon^{\lambda\nu\beta\gamma} P^\mu L_{\beta\gamma} + \epsilon^{\lambda\alpha\nu\gamma} P_\alpha L^\mu{}_\gamma + \epsilon^{\lambda\alpha\beta\nu} P_\alpha L_\beta{}^\mu - (\mu \leftrightarrow \nu)\right] \\ &= i\left(\eta^{\lambda\nu} W^\mu - \eta^{\lambda\mu} W^\nu\right) \iff [W^\lambda, L^{\mu\nu}] = i\left(W^\mu \eta^{\lambda\nu} - W^\nu \eta^{\lambda\mu}\right). \end{aligned} \tag{68}$$

3. **Proof that $C_2 = W_\mu W^\mu$ is a Casimir Operator (Eq. 4.132):**

$$\begin{aligned} [L^{\mu\nu}, W_\lambda W^\lambda] &= -i\eta_{\lambda\tau}\left((W^\mu \eta^{\lambda\nu} - W^\nu \eta^{\lambda\mu})W^\tau + W^\lambda (W^\mu \eta^{\tau\nu} - W^\nu \eta^{\tau\mu})\right) \\ &= -i\left(W^\mu W^\nu - W^\nu W^\mu + W^\nu W^\mu - W^\mu W^\nu\right) = 0. \end{aligned} \tag{69}$$

4. **Rest-Frame Evaluation of W^μ and C_2 (Eqs. 4.133 – 4.136):** In the rest frame $P^\mu = (m,0,0,0) \implies P_0 = -m, P_i = 0$. With $\epsilon^{0123} = -1$ ($\epsilon^{0ijk} = -\epsilon_{ijk}$) and $J_i = -\frac{1}{2}\epsilon_{ijk}L^{jk}$:

$$W^0 = 0, \qquad W^i = \frac{1}{2}\epsilon^{i0jk} P_0 L_{jk} = \frac{1}{2}\left(-\epsilon^{0ijk}\right)(-m)\left(-\epsilon_{jkl} J_l\right) = m J_i, \tag{70}$$

so $C_2 = \eta_{\mu\nu} W^\mu W^\nu = -(W^0)^2 + \delta_{ij} W^i W^j = m^2 \vec{J}^2 = m^2 s(s+1)$.

3.4 Massless Little Group $ISO(2)$ and Helicity (Eqs. 4.108 – 4.115, 4.137 – 4.149)

For the standard null reference momentum $k^\mu = (E, 0, 0, E)$, $k_\mu = (-E, 0, 0, E)$.

1. **Little Group Generators:** Requiring $\omega_{\rho\sigma}(l^{\rho\sigma})^\mu{}_\nu k^\nu = 2\omega^\mu{}_\sigma k^\sigma = 0$ with $\omega_{0i} = -\eta_i$ and $\omega_{ij} = \epsilon_{ijk}\theta_k$ gives $\eta_3 = 0$, $\theta_2 = -\eta_1$, and $\theta_1 = \eta_2$, so $\theta_i J_i + \eta_i K_i = \theta_3 J_3 + \eta_1(K_1 - J_2) + \eta_2(K_2 + J_1)$, spanned by:

$$J_3, \qquad N_1 = K_1 - J_2 = -\Pi_2, \qquad N_2 = K_2 + J_1 = \Pi_1. \tag{71}$$

2. **Commutation Relations (Eqs. 4.111, 4.139, 4.140):**

$$[J_3, \Pi_1] = [J_3, J_1 + K_2] = iJ_2 - iK_1 = i\Pi_2, \tag{72}$$
$$[J_3, \Pi_2] = [J_3, J_2 - K_1] = -iJ_1 - iK_2 = -i\Pi_1, \tag{73}$$
$$[\Pi_1, \Pi_2] = [J_1 + K_2, J_2 - K_1] = iJ_3 - iJ_3 = 0. \tag{74}$$

3. **Pauli-Lubanski Vector on Massless States (Eqs. 4.138, 4.145 – 4.147):** Evaluating $W^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}P_\nu L_{\rho\sigma}$ on $P_0 = -E, P_3 = E, P_1 = P_2 = 0$ (with $L_{ij} = -\epsilon_{ijk}J_k$ and $L_{0i} = K_i$):

$$W^0 = EJ_3, \qquad W^3 = EJ_3, \qquad W^1 = -E(J_1{+}K_2) = -E\Pi_1, \qquad W^2 = -E(J_2{-}K_1) = -E\Pi_2. \tag{75}$$

When $\Pi_1 = \Pi_2 = 0$ and $J_3|k,h\rangle = h|k,h\rangle$, we obtain $W^\mu = hP^\mu$ and $W_\mu W^\mu = h^2 P_\mu P^\mu = 0$.

---

4 Complete Solutions to Problems 4.1 and 4.2 (Section 4.2)

4.1 Problem 4.1: Decomposition of Rank-Two Lorentz Tensors and Massive Spin-2 Polarizations

(a) Decomposition of $(\frac{1}{2},\frac{1}{2}) \otimes (\frac{1}{2},\frac{1}{2})$

Using the $SU(2)$ Clebsch-Gordan decomposition $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ on each chiral factor:

$$\left(\frac{1}{2},\frac{1}{2}\right) \otimes \left(\frac{1}{2},\frac{1}{2}\right) = \left(\frac{1}{2} \otimes \frac{1}{2}, \frac{1}{2} \otimes \frac{1}{2}\right) = (0 \oplus 1, 0 \oplus 1) = (0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1). \tag{76}$$

Dimension check using $\dim(j_L, j_R) = (2j_L + 1)(2j_R + 1)$:

$$4 \times 4 = 16 = 1 + 3 + 3 + 9. \tag{77}$$

(b) Explicit Index Decomposition of $T^{\mu\nu}$

Define $T^{[\mu\nu]} = \frac{1}{2}(T^{\mu\nu} - T^{\nu\mu})$ and $T^{(\mu\nu)} = \frac{1}{2}(T^{\mu\nu} + T^{\nu\mu})$. Then:

$$T^{\mu\nu} = \underbrace{\frac{1}{4}\eta^{\mu\nu}T^\rho{}_\rho}_{(0,0),\ \dim=1} + \underbrace{T^{[\mu\nu]}}_{(1,0)\oplus(0,1),\ \dim=3+3=6} + \underbrace{\left(T^{(\mu\nu)} - \frac{1}{4}\eta^{\mu\nu}T^\rho{}_\rho\right)}_{(1,1),\ \dim=9}. \tag{78}$$

(c) Restriction to the Spatial Rotation Subgroup $SU(2)_{\text{rot}}$

Because $j_i = A_i + B_i$ generates the diagonal subgroup $SU(2)_{\text{rot}} \subset SU(2)_L \times SU(2)_R$, the representation (j_L, j_R) decomposes under spatial rotations via angular momentum addition:

$$(j_L, j_R)\big|_{SU(2)_{\text{rot}}} \simeq \bigoplus_{j=|j_L - j_R|}^{j_L + j_R} j. \tag{79}$$

- $(\frac{1}{2}, \frac{1}{2})\big|_{SU(2)_{\text{rot}}} = 0 \oplus 1$ (1 time component V^0 with $j = 0$, and 3 spatial components V^i with $j = 1$).
- $(1,0)\big|_{SU(2)_{\text{rot}}} = 1$ and $(0,1)\big|_{SU(2)_{\text{rot}}} = 1$ (self-dual and anti-self-dual combinations $\vec{E} \pm i\vec{B}$ each transform as 3-vectors of spin $j = 1$).
- $(1,1)\big|_{SU(2)_{\text{rot}}} = 0 \oplus 1 \oplus 2$ (S^{00} has $j = 0$ [1 component], S^{0i} has $j = 1$ [3 components], and traceless symmetric $S^{ij} - \frac{1}{3}\delta^{ij} S^{kk}$ has $j = 2$ [5 components]).

(d) Rest-Frame Analysis of Massive Spin-2 Polarization Tensor $\epsilon^{\mu\nu}(p)$

Given $\epsilon^{\mu\nu} = \epsilon^{\nu\mu}$, $\eta_{\mu\nu}\epsilon^{\mu\nu} = 0$, and $p_\mu \epsilon^{\mu\nu} = 0$ in the rest frame $p^\mu = (m, 0, 0, 0) \implies p_\mu = (-m, 0, 0, 0)$:

$$p_\mu \epsilon^{\mu\nu} = -m\epsilon^{0\nu} = 0 \implies \epsilon^{0\nu} = \epsilon^{\nu 0} = 0 \quad (\forall \nu \in \{0,1,2,3\}). \tag{80}$$

Only the 3×3 spatial submatrix ϵ^{ij} $(i, j \in \{1,2,3\})$ survives, subject to:

$$\epsilon^{ij} = \epsilon^{ji} \quad (6 \text{ components}), \qquad \eta_{\mu\nu}\epsilon^{\mu\nu} = -\epsilon^{00} + \delta_{ij}\epsilon^{ij} = \delta_{ij}\epsilon^{ij} = 0 \quad (-1 \text{ constraint}), \tag{81}$$

leaving $6 - 1 = 5$ independent components.

(e) Spin-2 Irreducible Representation Under $SO(3)$

Under the rest-frame little group $SO(3)$, a general spatial tensor t^{ij} in $\mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$ $(1 \otimes 1 = 0 \oplus 1 \oplus 2)$ decomposes into trace ($\mathbf{1}$, $j = 0$), antisymmetric ($\mathbf{3}$, $j = 1$), and symmetric traceless ($\mathbf{5}$, $j = 2$) parts. Since ϵ^{ij} is symmetric and traceless, it transforms irreducibly in the $2(2) + 1 = 5$-dimensional spin-2 ($j = 2$) representation of $SO(3)$, describing the 5 polarization states of a massive spin-2 particle.

4.2 Problem 4.2: Explicit Index Proof of the Fierz Identity (Eqs. 4.150 – 4.151)

Step 1: Completeness Relation for $(\sigma^\mu)_{\alpha\dot{\alpha}}(\sigma_\mu)_{\beta\dot{\beta}}$ in $(-+++)$ Signature with $\epsilon_{\dot{\alpha}\dot{\beta}} = -\epsilon_{\alpha\beta}$
Using $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ and $\sigma^\mu = (\mathbf{1}, \sigma^1, \sigma^2, \sigma^3)$:

$$(\sigma^\mu)_{\alpha\dot{\alpha}}(\sigma_\mu)_{\beta\dot{\beta}} = -(\sigma^0)_{\alpha\dot{\alpha}}(\sigma^0)_{\beta\dot{\beta}} + (\sigma^1)_{\alpha\dot{\alpha}}(\sigma^1)_{\beta\dot{\beta}} + (\sigma^2)_{\alpha\dot{\alpha}}(\sigma^2)_{\beta\dot{\beta}} + (\sigma^3)_{\alpha\dot{\alpha}}(\sigma^3)_{\beta\dot{\beta}}. \tag{82}$$

Checking components against $\epsilon_{12} = +1, \epsilon_{21} = -1$ and the professor's convention $\epsilon_{\dot{1}\dot{2}} = -1, \epsilon_{\dot{2}\dot{1}} = +1$:

- $(\alpha, \dot{\alpha}, \beta, \dot{\beta}) = (1,1,1,1)$: $-1 + 0 + 0 + 1 = 0 = +2\epsilon_{11}\epsilon_{\dot{1}\dot{1}}$.
- $(\alpha, \dot{\alpha}, \beta, \dot{\beta}) = (1,1,2,2)$: $-1 + 0 + 0 - 1 = -2 = +2\epsilon_{12}\epsilon_{\dot{1}\dot{2}}$ (since $\epsilon_{12}\epsilon_{\dot{1}\dot{2}} = (+1)(-1) = -1$).
- $(\alpha, \dot{\alpha}, \beta, \dot{\beta}) = (1,2,2,1)$: $-0 + 1 + (-i)(i) + 0 = +2 = +2\epsilon_{12}\epsilon_{\dot{2}\dot{1}}$ (since $\epsilon_{12}\epsilon_{\dot{2}\dot{1}} = (+1)(+1) = +1$).

- $(\alpha, \dot{\alpha}, \beta, \dot{\beta}) = (1, 2, 1, 2)$: $-0 + 1 + (-i)^2 + 0 = 0 = +2\epsilon_{11}\epsilon_{\dot{2}\dot{2}}$.

By symmetry under $1 \leftrightarrow 2$, this proves the completeness identity for all index values:

$$(\sigma^\mu)_{\alpha\dot{\alpha}}(\sigma_\mu)_{\beta\dot{\beta}} = +2\epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}}. \tag{83}$$

Step 2: Contraction with Spinors With $\chi\sigma^\mu\bar{\psi} = \chi^\alpha(\sigma^\mu)_{\alpha\dot{\alpha}}\bar{\psi}^{\dot{\alpha}}$ and $\lambda\sigma_\mu\bar{\rho} = \lambda^\beta(\sigma_\mu)_{\beta\dot{\beta}}\bar{\rho}^{\dot{\beta}}$, and using $\lambda_\alpha = \epsilon_{\alpha\beta}\lambda^\beta$ and $\bar{\psi}_{\dot{\beta}} = \epsilon_{\dot{\beta}\dot{\alpha}}\bar{\psi}^{\dot{\alpha}} = -\epsilon_{\dot{\alpha}\dot{\beta}}\bar{\psi}^{\dot{\alpha}}$:

1. **For Commuting (c-number) Spinors ($\bar{\psi}^{\dot{\alpha}}\lambda^\beta = +\lambda^\beta\bar{\psi}^{\dot{\alpha}}$):**

$$\begin{aligned}(\chi\sigma^\mu\bar{\psi})(\lambda\sigma_\mu\bar{\rho}) &= (\sigma^\mu)_{\alpha\dot{\alpha}}(\sigma_\mu)_{\beta\dot{\beta}}\chi^\alpha\bar{\psi}^{\dot{\alpha}}\lambda^\beta\bar{\rho}^{\dot{\beta}} \\ &= +2\epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}}\chi^\alpha\lambda^\beta\bar{\psi}^{\dot{\alpha}}\bar{\rho}^{\dot{\beta}} \\ &= +2\left(\chi^\alpha\epsilon_{\alpha\beta}\lambda^\beta\right)\left(-\epsilon_{\dot{\beta}\dot{\alpha}}\bar{\psi}^{\dot{\alpha}}\bar{\rho}^{\dot{\beta}}\right) \\ &= -2\left(\chi^\alpha\lambda_\alpha\right)\left(\bar{\psi}_{\dot{\beta}}\bar{\rho}^{\dot{\beta}}\right) = -2(\chi\lambda)(\bar{\psi}\bar{\rho}).\end{aligned} \tag{84}$$

2. **For Anticommuting (Grassmann) Spinors ($\bar{\psi}^{\dot{\alpha}}\lambda^\beta = -\lambda^\beta\bar{\psi}^{\dot{\alpha}}$):** Reordering $\bar{\psi}^{\dot{\alpha}}$ past λ^β introduces an extra minus sign, yielding $+2(\chi\lambda)(\bar{\psi}\bar{\rho})$ (or $-2(\chi\lambda)(\bar{\psi}\bar{\rho})$ if the dotted contraction is defined as $\bar{\psi}\bar{\rho} = \bar{\psi}^{\dot{\alpha}}\bar{\rho}_{\dot{\alpha}}$).